

ESERCIZIO 1:

a) $\forall \varepsilon > 0, \exists \delta > 0$ t.c. $|f(x) - 3| < \varepsilon \quad \forall x$ t.c. $0 < |x - 4| < \delta$

b) **TEO (ROLLE):** $a, b \in \mathbb{R}, a < b, f: [a, b] \rightarrow \mathbb{R}$, SUPPONIAMO CHE:

1) f È CONTINUA SU $[a, b]$ 2) f È DERIVABILE SU (a, b) 3) $f(a) = f(b)$

ALLORA $\exists \hat{x} \in (a, b)$ t.c. $f'(\hat{x}) = 0$.

DIMOSTRAZIONE:

CASO 1: f È COSTANTE, QUINDI $f' = 0 \quad \forall x \in (a, b)$

CASO 2: f NON È COSTANTE, ALLORA:

PER WEIERSTRASS, $\exists x_m, x_M \in [a, b]$ t.c. $f(x_m) = \min f < \max f = f(x_M)$

QUINDI $x_m \vee x_M \notin \{a, b\}$ OUNERO $x_m \vee x_M \in (a, b)$, QUINDI PER FERMAT
 $f'(x_m) \vee f'(x_M) \Rightarrow$ SONO UGUALI A ZERO.

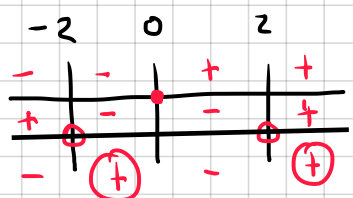
ESERCIZIO 2:

$$f(x) = \frac{x^3}{x^2 - 4}$$

DOMINIO: $x^2 - 4 \neq 0 \Rightarrow x \neq \pm 2$ QUINDI $x \in (-\infty, -2) \cup (-2, 2) \cup (2, +\infty)$
 $\mathbb{R} \setminus \{-2, +2\}$

SIMMETRIE: $f(-x) = \frac{(-x)^3}{(-x)^2 - 4} = -\frac{x^3}{x^2 - 4} = -f(x)$ QUINDI È DISPARI
 (SIMMETRIA RISPETTO L'ORIGINE)

SEGNO: $f(x) \geq 0 \quad \frac{x^3}{x^2 - 4} \geq 0$ $F_1 \geq 0 \quad x^3 \geq 0 \quad x \geq 0$
 $F_2 > 0 \quad x^2 - 4 > 0 \quad x < -2 \vee x > 2$



$$f(x) > 0 \quad x \in (-2, 0) \cup (2, +\infty)$$

$$f(x) = 0 \quad x = 0$$

$$f(x) < 0 \quad x \in (-\infty, -2) \cup (0, 2)$$

LIMITI: $\lim_{x \rightarrow -2^+} \frac{x^3}{x^2 - 4} = -\infty$ $\lim_{x \rightarrow -2^-} \frac{x^3}{x^2 - 4} = +\infty$

$$\lim_{x \rightarrow 2^+} \frac{x^3}{x^2 - 4} = +\infty \quad \lim_{x \rightarrow 2^-} \frac{x^3}{x^2 - 4} = -\infty$$

NO ASINTOTI
ORIZZONTALI

$$\lim_{x \rightarrow +\infty} \frac{x^3}{x^2 - 4} = \lim_{x \rightarrow +\infty} \frac{x^3}{x^2(1 - \frac{4}{x^2})} = +\infty \quad \lim_{x \rightarrow -\infty} \frac{x^3}{x^2 - 4} = -\infty$$

VERIFICHIAMO EVENTUALI ASINTOTO OBLIQUI:

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \lim_{x \rightarrow +\infty} \frac{x^3}{x^3 - 4x} = \lim_{x \rightarrow +\infty} \frac{x^3}{x^3(1 - \frac{4}{x^2})} =$$

$$= 1 = m \quad \lim_{x \rightarrow +\infty} f(x) - (1x) = \lim_{x \rightarrow +\infty} \frac{x^3}{x^2 - 4} - x = \lim_{x \rightarrow +\infty} \frac{x^3 - x^3 - 4x}{x^2 - 4} =$$

$$= \lim_{x \rightarrow +\infty} \frac{-4x}{x^2 - 4} = \lim_{x \rightarrow +\infty} \frac{-4x}{x^2(1 - \frac{4}{x^2})} = 0 \quad y = x$$

$\hookrightarrow 0$

f HA UN ASINTOTO OBLIQUO PER $x \rightarrow +\infty$ CON EQUAZIONE $y = x$ E PER SIMMETRIA ANCHE PER $x \rightarrow -\infty$.

DERIVATA PRIMA:

$$f'(x) = \left(\frac{x^3}{x^2 - 4} \right)' = \frac{3x^2(x^2 - 4) - x^3(2x)}{(x^2 - 4)^2} =$$

$$= \frac{3x^4 - 12x^2 - 2x^4}{(x^2 - 4)^2} = \frac{x^4 - 12x^2}{(x^2 - 4)^2} = \frac{x^2(x^2 - 12)}{(x^2 - 4)^2}$$

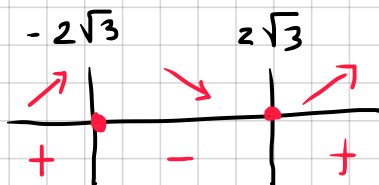
È DERIVABILE SUL SUO DOMINIO.

MONOTONIA: $f'(x) > 0 \Leftrightarrow \frac{x^2(x^2 - 12)}{(x^2 - 4)^2} > 0$ QUINDI $x^2 - 12 > 0$

$\hookrightarrow > 0 \forall x$

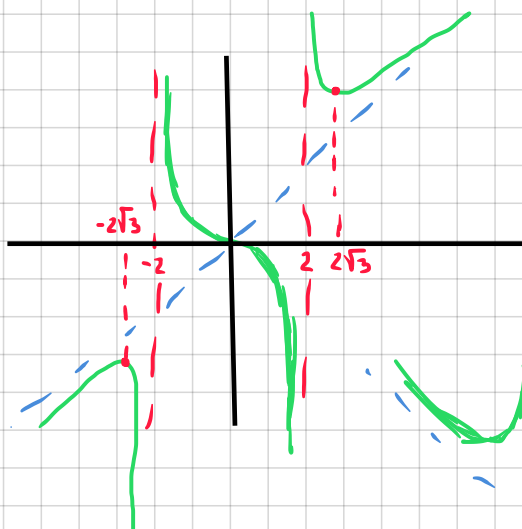
$$x < -\sqrt{12} \vee x > +\sqrt{12} \Leftrightarrow x < -2\sqrt{3} \vee x > 2\sqrt{3}$$

QUINDI: $f'(x)$



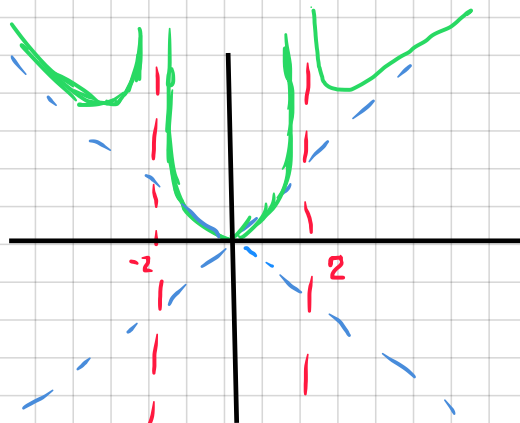
MIN LOCALE MAX LOCALE
 $\downarrow$ $\downarrow$
 $x = -2\sqrt{3}$ E $x = 2\sqrt{3}$ SONO
 PUNTI DI ESTREMO LOCALE.

GRAFICO:



$$g(x) = |f(x)|$$

$\Rightarrow$ GRAFICO



ESERCIZIO 3: $\lim_{x \rightarrow 0} \frac{\tan(\sin(x))}{\log(1+3x)} =$ $\log(1+3x) \sim 3x \quad \forall x \rightarrow 0$
 $\tan(\sin(x)) \sim \sin(x) \quad \forall x \rightarrow 0$
 $\sin(x) \sim x \quad \forall x \rightarrow 0$

$$= \lim_{x \rightarrow 0} \frac{x}{3x} = \frac{1}{3}$$

$$\lim_{x \rightarrow +\infty} \frac{\log(x) - 6x^4 + x^3}{e^{-x} + 2x^4} = \lim_{x \rightarrow +\infty} \frac{x^4 \left(\frac{\log(x)}{x^4} - 6 + \frac{1}{x} \right)}{x^4 \left(\frac{e^{-x}}{x^4} + 2 \right)} = \frac{-6}{2} = -3$$

ESERCIZIO 4:

a) $\int \frac{x^{-1}}{x^2-2x-3} dx = \frac{1}{2} \int \frac{2x-2}{x^2-2x-3} dx = \frac{1}{2} \log|x^2-2x-3| + c$

b) $\int x^2 e^{\frac{x}{2}} dx = 2x^2 e^{\frac{x}{2}} - \int 4x e^{\frac{x}{2}} dx =$

$$= 2x^2 e^{\frac{x}{2}} - \left(8x e^{\frac{x}{2}} - \int 8e^{\frac{x}{2}} dx \right) =$$

$$= 2x^2 e^{\frac{x}{2}} - 8x e^{\frac{x}{2}} - 8 \int e^{\frac{x}{2}} dx =$$

$$= 2x^2 e^{\frac{x}{2}} - 8x e^{\frac{x}{2}} - 16e^{\frac{x}{2}} + c =$$

$$= e^{\frac{x}{2}} (2x^2 - 8x - 16) + c = 2e^{\frac{x}{2}} (x^2 - 4x - 8) + c$$

$$\begin{array}{rcl} \int & & \\ 2x & - & x^2 \\ e^{\frac{x}{2}} & \rightarrow & 2e^{\frac{x}{2}} \\ 4 & - & 4x \\ e^{\frac{x}{2}} & \rightarrow & 2e^{\frac{x}{2}} \end{array}$$